

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Comparative Analysis

Code of Conduct:

I P Jyothika Reddy(192311156) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	

Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model True Positives False Positives False Negatives

A (Viola-Jones) 80 20 30

B (YOLO) 90 15 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

ANSWER:

1. Detection Performance Comparison

Formula:

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$

Model A – Viola-Jones

Precision = $80 / (80 + 20) = 0.80 = 80\%$

Recall = $80 / (80 + 30) = 0.727 = 72.73\%$

Model B – YOLO

Precision = $90 / (90 + 15) = 0.857 = 85.71\%$

Recall = $90 / (90 + 20) = 0.818 = 81.82\%$

Comparison: YOLO has higher precision (85.71%) and recall (81.82%) than Viola-Jones (80% and 72.73%).

Conclusion: YOLO is better for object detection because it detects more actual objects and produces fewer false detections.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

FrameMethod A Method B

1 2 3

2 3 5

3 2 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

ANSWER:

2. Motion Estimation Error Analysis

Method A

Errors = 2, 3, 2

Average error = $(2 + 3 + 2) / 3$
 $= 7/3 = 2.33$ pixels

Method B

Errors = 3, 5, 4

Average error = $(3 + 5 + 4) / 3$
 $= 12/3 = 4$ pixels

Comparison: Method A has an average error of 2.33 pixels, while Method B has 4 pixels.

Conclusion: Method A is more accurate because it has the lower average displacement error.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1 5 8

R2 6 10

R3 5 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

ANSWER:

Optical Flow Magnitude Comparison

Method A

Average = $(5 + 6 + 5) / 3$

$$= 16/3 = 5.33 \text{ pixels/frame}$$

Method B

$$\begin{aligned}\text{Average} &= (8 + 10 + 9) / 3 \\ &= 27/3 = 9 \text{ pixels/frame}\end{aligned}$$

Comparison: Method B has a higher average motion magnitude (9 pixels/frame) than Method A (5.33 pixels/frame).

Conclusion: Method B captures stronger motion because it produces a higher average optical-flow magnitude.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

ANSWER:

Depth Estimation Comparison

Stereo Vision

Actual depths: 2, 3, 4 m

Estimated: 2, 3, 4 m

Absolute errors:

- A: $|2 - 2| = 0 \text{ m}$
- B: $|3 - 3| = 0 \text{ m}$
- C: $|4 - 4| = 0 \text{ m}$

Average absolute error = 0 m

Structure from Motion (SfM)

Estimated: 2.5, 3.8, 5.0 m

- A: $|2.5 - 2| = 0.5 \text{ m}$
- B: $|3.8 - 3| = 0.8 \text{ m}$

- C: $|5.0 - 4| = 1.0$ m

Average absolute error:

$$(0.5 + 0.8 + 1.0) / 3 \\ = 2.3/3 = 0.77 \text{ m}$$

Comparison: Stereo Vision has 0 m average error, while SfM has 0.77 m average error.

Conclusion: Stereo Vision is more accurate because its estimated depths exactly match the actual depths.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

ANSWER:

Pattern Recognition Accuracy Comparison

Formula:

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$$

Traditional System

$$\text{Accuracy} = (85 / 100) \times 100 \\ = \mathbf{85\%}$$

Deep Learning

$$\text{Accuracy} = (92 / 100) \times 100 \\ = \mathbf{92\%}$$

Comparison: Deep Learning achieves **92%**, while the Traditional system achieves **85%**.

Conclusion: Deep Learning performs better with an accuracy improvement of **7 percentage points**.